

Evaluation of the waste management system at the Züri Fäscht

Bachelor Thesis

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Evaluation of the waste management system at the Züri Fäscht

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Bachelor Thesis

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Abstract

This thesis evaluates the waste management system implemented for Züri Fäscht, the biggest festival in Zurich. It aims to identify opportunities for optimization in environmental impact reductions and resource efficiency. The study encompasses the analysis of the material flow, transportation, costs, energy, and emissions. Comparisons are made between the latest editions with other big events worldwide.

Sorting is a new waste treatment strategy adapted by the festival organizers. It caused a lot of logistical complications but showed that not many recyclables are left in the waste and above all, showed that sorting at source has a major impact on the event. The earlier the recyclables are sorted, the better they can be recovered and remain in the life cycle. The environmental impact was difficult to evaluate due to imprecise data.

Ultimately, the work contributes to the discourse on sustainable event management by recording the aftermath, providing practical recommendations, informing organizers, and raising awareness among visitors.

Contents

1	Introduction	1
2	Theory and Background	3
2.1	Municipal Solid Waste	3
2.2	Life Cycle	3
2.3	Waste Management System	4
2.3.1	Waste Collection & Sorting	5
2.3.2	Waste treatment & disposal	6
2.4	ERZ	7
2.5	City festival: Züri Fäscht	8
2.6	Problem Definition	8
3	Methodology	10
3.1	Study Area	10
3.1.1	During the festival	10
3.1.2	After the festival	13
3.2	Qualitative Methods	13
3.2.1	Interviews	13
3.2.2	Individual experience	14
3.2.3	Festivals in comparison	15
3.3	Quantitative Methods	16
3.3.1	Material flow analysis	16
3.3.2	Transport	17
3.3.3	Costs	17
3.3.4	Energy and emissions	18
4	Results and Discussion	20
4.1	Material flow analysis	20
4.2	Transport	24
4.3	Costs	26

4.4	Energy and emissions.....	27
4.5	Festivals in comparison.....	30
5	Conclusion.....	35
6	Outlook.....	37
7	References	38
	Appendix.....	42

List of Figures

Figure 1. Schematic representation of circular economy.....	4
Figure 2. Waste hierarchy.....	5
Figure 3. Waste management system adopted for Züri Fäscht (Züri Fäscht, 2023a).....	10
Figure 4. Disposal points from Züri Fäscht 2019 (left) and 2023 (right).....	11
Figure 5. Layout of the sectors in the city (left) and a detailed view of area 2B (right).	12
Figure 6. Sankey diagram for waste collected from Züri Fäscht 2019 in tonnes.....	20
Figure 7. Sankey diagram for waste collected from Züri Fäscht 2023 in tonnes.....	21
Figure 8. Materials saved from incineration at the different sorting facilities.....	22
Figure 9. CED in comparison.	28
Figure 10. GTP100 in comparison.....	28
Figure 11. CED of different materials.	29
Figure 12. GTP100 of different materials.....	30

List of Tables

Table 1. Characterization of different festivals.....	16
Table 2. Material recovery rate.	23
Table 3. Routes of ZF 2019.	25
Table 4. Routes of ZF 2023.	25
Table 5. Distance per kilogram waste collected.....	26
Table 6. Total costs of the WMS during the festival.	27
Table 7. Impact avoided through recycling.	30
Table 8. General overview of Swiss festivals.....	32
Table 9. Information source for festivals compared in this work.	42
Table 10. Addresses used for transport calculations.	42
Table 11. Information sources for estimating the energy and emissions.	43

List of abbreviations

CED	Cumulative Energy Demand	
EP	Disposal point	de.: Entsorgungspunkt
ERZ	Entsorgung + Recycling Zürich	eng.: disposal recycling Zurich
GHG	Greenhouse Gas	
GTP	Global Temperature Potential	
LCA	Life Cycle Assessment	
LP	Logistics point	de.: Logistikpunkt
MFA	Material Flow Analysis	
MRF	Material Recovery Facility	
MRR	Material Recovery Rate	
MSW	Municipal Solid Waste	
PET	Polyethylene Terephthalate	
WMS	Waste Management System	
WtE	Waste to energy plant	
ZWI	Zero Waste Index	

1 Introduction

Our population is growing and so is the waste production. Every year, 2 billion tonnes of solid waste are produced worldwide. Countries with a high-income level and high urbanization rate produce the most (Kaza et al., 2021). The rising waste production causes more greenhouse gases (GHG) to get captured in the atmosphere and contributes significantly to the overall global warming (Chandrappa & Das, 2012). The Paris Agreement 2015 is a significant step to acknowledge the climate change and to set a common goal among the nations. In order to reach the climate neutral state, every nation must take action (UNFCCC, 2015).

Switzerland produces around 6 million tons per year of municipal solid waste and takes one of the top places among the European countries. The amount of waste is positively correlated with GDP and, therefore, with the standard of living. With a population of 8.7 million, each Swiss citizen generates more than 700 kg of municipal solid waste per year, far above the world's 270 kg average (The World Bank Group, 2023) . In Switzerland, around half of it is recycled; the rest is incinerated (UFAM, 2022). A developed country, like Switzerland, that produces a big amount of waste is obliged to bear the responsibility. Furthermore, solving the problem with all the possessed resources and know-how, offers a possibility to inspire other actors.

In a world where the environmental problems are continually growing, it is important to take responsibility in every area of life. Flying, livestock farming, or deforestation are known factors to increasing the CO₂ pollution. There are many areas that, at first glance, do not look like an environmental sin, yet they do contribute. This thesis looks at the ecological repercussions caused by festivals and major events, shedding light on their environmental impact. The focus is mainly put on waste management of the city festival Züri Fäscht and the city's attempt towards the eco-friendly waste treatment. The waste management is a critical point and the most expensive part of the waste cycle. Therefore, it needs planning, collecting and analysing data, efficient execution, and adequate technology.

The event organizers focus on minimizing waste through sorting, recycling, and reducing the single-use plastics. By embracing the lessons learned from these events, we can

collectively work towards a more environmentally responsible and sustainable world, where our celebrations harmonize with the planet's well-being.

2 Theory and Background

In this chapter, important terms and processes, relevant to this thesis, are described. The chapter begins with the definition of municipal solid waste and waste management. Then, a Life Cycle and sustainable system terms are introduced. Afterwards, the different stages of garbage collection, resorting, treatment, and disposal are discussed. In the end, the roles of ERZ, the organization managing waste in Zurich, and Züri Fäscht, the city festival producing waste responsibly, are described.

2.1 Municipal Solid Waste

Municipal solid waste, or MSW, is the umbrella term of everyday items that people throw away after use. This can be packaging, newspapers, leftovers, empty bottles, batteries and so forth. Statistics are often hard to evaluate and compare because of the different definitions of MSW and its heterogeneity. Generally, it consists of household, institutional, and commercial waste, and accounts for 3% of the solid waste (McDougall et al., 2008).

This work focuses on the waste that is thrown away into the public bins or on the ground. A distinction is made between glass, cardboard, aluminium, polyethylene terephthalate (PET), and residual waste. The correct treatment of MSW is crucial to minimize the environmental impact and to protect the nature. The main objectives towards safe waste management are pollution minimization and resource conservation. There are two steps for managing the municipal solid waste. First, the amount of generated waste needs to be reduced. Second, waste needs to be disposed of correctly and recovered efficiently (McDougall et al., 2008). The focus of this work is only on the second step, correct collection and disposal of MSW.

2.2 Life Cycle

Circular economy and linear economy are two contrasting approaches to managing resources and waste within an economic system. Waste generation is a significant outcome of a linear economy and the currently applied economic model worldwide. Products and materials, that are no longer useful, are often sent to landfills or incinerated, contributing to environmental pollution and resource depletion. In a circular economy, waste is minimized through practices such as recycling, reusing, upcycling, and waste-to-energy processes. The aim of circular economy is to keep materials and products in use as long as possible, reducing the need for incineration. Instead of removing the utensil, it is repurposed and stays in the

recycling flow. Figure 1 shows the concept of circular economy, where the waste is a fed back into the industry and the resources are maintained (Liu & Ramakrishna, 2021).

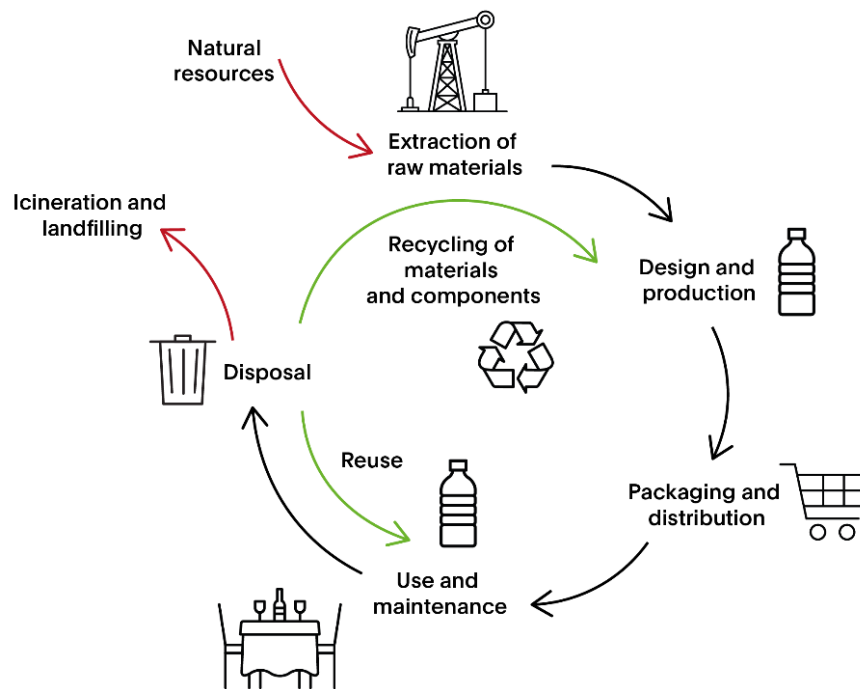


Figure 1. Schematic representation of circular economy.

Life Cycle Assessment (LCA) is a method, used in circular economy, to record and review the environmental impact of products and processes. It quantifies the emissions of a service from its production and use until its end of life. LCA ranks the one-time use of products low and favours the circular approach (Liu & Ramakrishna, 2021). It calculates the overall environmental impact of all stages together, starting with the raw material and ending with the final disposal. LCA allows to connect various actors in circular economy: the manufacturers, the waste collection and disposal system (Pires et al., 2019).

2.3 Waste Management System

Every Waste Management System should ensure that the waste generated at source is properly processed. Its main components are: waste production, waste collection and transport, waste treatment and reprocessing, and its final disposition. Reducing waste should be the system's objective and top priority (Demirbas, 2011).

The zero waste concept is the idealistic philosophy where products create no waste (Zaman, 2014). If a material faces the end-of-life it gets redirected into the manufacturing process and ideally never leaves the cycle, therefore, produces no harmful waste. The R

principles - Reduce, Reuse, Recycle, Recover, are a helpful tool to implement the concept. Waste management is engaged in identifying the zero waste indicators and finding solutions concerning sustainability (Zaman, 2014).

The waste hierarchy, schematically represented in Figure 2, is a prioritization framework for managing waste, designed to minimize the environmental impact of waste generation and disposal. The waste management system follows a hierarchical order of preference, with the most preferred option at the top and the least preferred one at the bottom. The concept is often adapted to the situation and comes in different versions (Zhang et al., 2022).

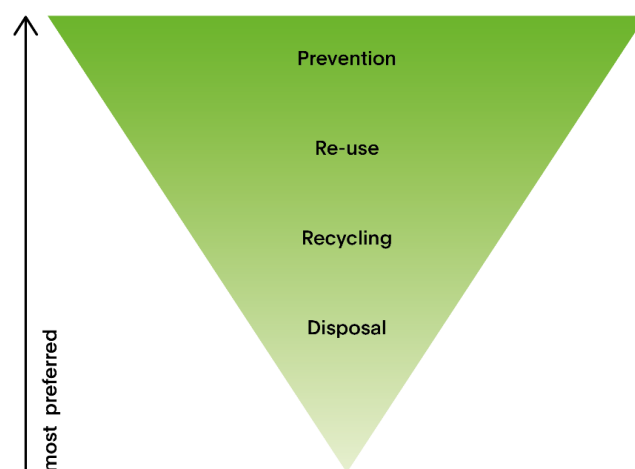


Figure 2. Waste hierarchy.

2.3.1 Waste Collection & Sorting

Collecting waste is one of the most expensive steps in the cycle. Waste collection is the link between the waste producers and the waste processors and determines the implementation of the following tasks and their success. The generators want a convenient and timesaving way to get rid of the waste, while the fabricators rely on a well sorted quantity for optimal treatment. The challenge is to find a suitable intersection. Therefore, the MSW system targets time savings, cost optimization, and good logistics as steps on a way to efficiency (Pires et al., 2019).

Waste collection of commercial waste is based on different strategies. It can be a “bring system” where people go to a sorting station and are themselves responsible for correct sorting. Cities are increasingly installing segregation facilities. Another, more consumer convenient solution, are the public bins with no at-source material sorting. Depending on the solution different tools are needed. Vacuum sweeper and brooms are the most used utensils for waste

from the ground collection. In every scenario, the final collection step is the transfer to the disposal or processing facilities (Han & Cueto, 2015).

Resorting comes after collection and before disposal. It is a helpful way to reduce contamination and save recyclables. Contamination, or impurity, includes dirt or incorrectly separated items. Depending on the material, the tolerance of impurity can vary. Resorting MSW includes various process steps that depend on the facility. There can be magnets, hand sorters, X-ray sensors and many more (Chandrappa & Das, 2012). After resorting the separated materials go different ways, i.e., recycling, downcycling, or incineration, depending on their potential, with a goal to reuse them in their best possible way.

2.3.2 Waste treatment & disposal

Depending on the collected and resorted material, there are different procedures of recirculation. Some waste treatment and disposal procedures can save more material than others and, therefore, contribute more to the life cycle of a good. The most common and the most important processes for this thesis are explained below.

Reprocessing: Recycling & Downcycling

Recycling is a well-known step in the waste management procedure. Its goal is the energy and material recovery. It prevents using new raw material, saves emissions, and energy. It is resource-efficient and extends the life of products and materials (Chandrappa & Das, 2012). Most of materials in the MSW can be recycled, if contamination is kept within technological limits, depending on the susceptibility of the material. Carboard, glass and aluminium are easily recyclable without losing properties. Plastic, on the other hand, is more difficult, because of various possible compositions, with PET being the most suitable one for circular recycling (Voort, 2018). Downcycling takes place where the reprocessed material cannot maintain the same properties throughout the Life Cycle and the product loses a part of its value (Pires et al., 2019). Consequently, there is a reduction of quality and lower functionality compared to the original state (Helbig et al., 2022).

Incineration and Waste to Energy

Incineration is a thermal treatment for MSW. It is a convenient way to reduce, recover and sanitize waste. Heat and electricity are the final benefits of the process. The produced ash can be used in the construction or metal industry. The negative side are the emissions and the residual product, the ash. It contains harmful pollutants as dioxins, acid gases, nitrogen oxide, heavy metal, and particulates that can get distributed into the air (McDougall et al., 2008). In

Switzerland a little less than half of the MSW is incinerated, 2.5% of the nationwide energy requirement and 4% of the electricity production (FOEN, 2022).

A waste-to-energy (WtE) plant burns waste to generate electricity and heat. The primary benefit of the process is to recover energy from waste that would otherwise be landfilled. It has the potential to diminish the environmental impact, contingent on what it substitutes. For instance, a WtE plant may have a large positive impact, if it replaces the heat production from coal; it can also be less beneficial if it replaces the biofuel (Tyskeng Sara, 2010).

Landfilling

The storage of the residual waste is long term. Along the way, the decomposition process is not supervised. Landfills are the least favourable waste disposal option. However, they are one of the cheapest and easiest ways to handle waste (Chandrappa & Das, 2012). The emissions of landfills are leachate and gases, both of which must be captured to avoid contaminating the ground water and the air. Additionally, public acceptance and local suitability are often the biggest challenges when looking for new landfilling stations (McDougall et al., 2008). In Switzerland, the municipal waste is not landfilled at all, therefore, landfills will not be further elaborated in this work (FDFA, 2023).

2.4 ERZ

ERZ (German: *Entsorgung + Recycling Zürich*) is the waste management responsible in Zurich. The head office is on the outskirts of the city, in Hagenholz, where the waste recycling plant operates to cogenerate the CO₂-neutral electricity and heat (Stadt Zürich, 2023b).

Their main ERZ's responsibilities are keeping the public space clean, managing the effluent and treat the waste. ERZ manages the waste generated by the society in an environmentally responsible and sustainable manner, with the goal of minimizing the environmental impacts, conserving resources, and promoting a cleaner environment. Prevention work through awareness campaigns and educational work are also parts of it. The circular economy is the company's fundamental philosophy and it influences all their planning and actions. Trade-offs are being made between material savings, cost savings, waste reduction, market opportunities, regulatory compliance, and innovation. Managing solid waste is more than just a community service, there is a whole business industry behind it (Stadt Zürich, 2023c).

2.5 City festival: Züri Fäscht

Züri Fäscht is a festival that takes place every three years and occurs in the city center of Zurich. For three days, visitors can experience versatile entertainment while enjoying the scenes of Zurich. The first edition, a small summer folk festival, took place in 1951. Today the festival counts more than 2 million visitors and is more than just a local spectacle. The event features music, theatre, dance, and visual art performances at venues located primarily around the lake and the river basin. With the size of the festival, its reach also increased. It is an important part of the Zurich tourism industry and an advertisement for Switzerland. Viewers from all around the world come to participate and to experience the city's culture (Züri Fäscht, 2023a).

2.6 Problem Definition

Züri Fäscht generates a lot of waste in a short amount of time, which ERZ manages. This challenge encompasses many aspects. First, it is expected that the trash and disorder on the streets will be cleaned by the next day. Major transportation axes need to be usable, so that traffic can move smoothly. The residents expect the usual cleanliness and there is an expectation for an environmentally responsible approach from the festival participants and organizers. The municipal solid waste may still contain valuable resources, that can be recovered, reused, and repurposed. The more items is saved, the more the volume of waste is reduced. This approach contributes to the environmental protection and transition to a circular economy. In summary, the goal of this thesis is to evaluate the waste management concept adapted by the festival organizers and ERZ, according to the criteria of environmental friendliness and feasibility with given operating resources.

The previous edition of Züri Fäscht took place back in 2019. Since then, the awareness surrounding the imperative for festivals to adopt the environmentally conscious practices has gained significance. Sustainability became a business and opened new marketing strategies. The organizers did also implement the first climate friendly steps, but there was still no official sustainability concept. After that, the desire and need came to plan for the upcoming Züri Fäscht in 2023. ERZ itself took over the waste management planning. New approaches, such as pre-sorting, recycling stations, more frequent garbage collection, and depot system, were introduced. The question arose on whether and how sustainable these new strategies are. This thesis conducts an assessment to examine the environmental impacts of sustainable practices

in the festival's waste management system. It also examines the feasibility of the concept with the given resources, such as the logistics and the costs.

3 Methodology

This chapter describes different approaches for waste management at Züri Fäscht and evaluates the environmental impact of this and previous editions of the festival.

Qualitative data was collected, the comparison has been made with other festivals and events through interviews and online research. The observation has been made and the individual experience was reflected and evaluated. The fuel and energy consumption for transportation routes were monitored, the material flows were validated, and the emissions were calculated. In the first part of the, the study area was described, enhancing the understanding of the project and its surroundings.

3.1 Study Area

3.1.1 During the festival

Züri Fäscht took place primarily around the lake basin and the Limmat river. The non-gated area was open for everyone, independent on the age. Families with children came, as well as the senior citizens, young couples, and individuals who resided and were employed in the area. At night the participants were mainly young adults, and adolescents. Around 200 feast stands and 320 market stands provided the visitors with food, beverages, and souvenirs. This infrastructure, which influenced the waste stream, is illustrated in Figure 3. The sustainability strategy of festival is represented for waste collection and recycling

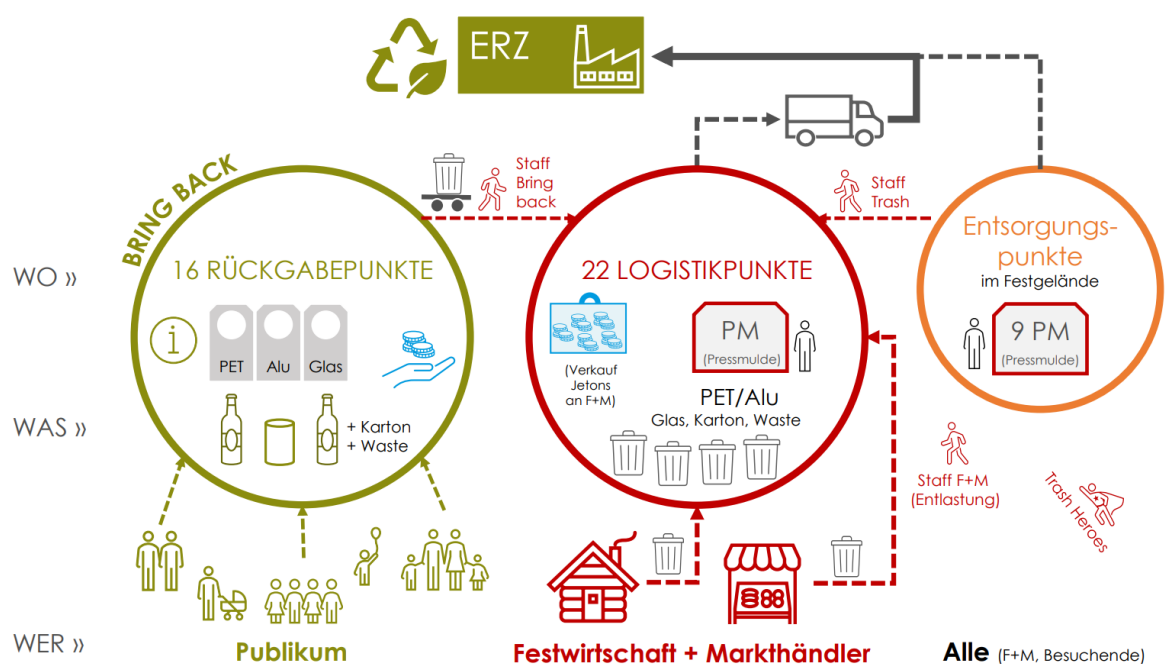


Figure 3. Waste management system adopted for Züri Fäscht (Züri Fäscht, 2023a).

The audience brought back the recyclables, such as the PET cups, the aluminium cans, and the glass bottles, to the return points, together with a jeton and receive the 2 CHF deposit. The visitors could bring objects bought in superstores to the disposal points (EP). There were more than 130 EP in the entire area, all built over the period of two days before the festival. An EP was planned on a squared or triangular area and was enclosed with metal grids. The green colours were intended to make the location clearly visible, and the pictograms helped to dispose of the waste correctly. Figure 4 gives a comparison between the disposal points in 2019 and 2023. In 2019 the black containers were strung together in a line. The sorting at source stations were also marked with a green sign, aluminium and PET were collected in the same container and there were no barriers between the visitors and the disposal.



Figure 4. Disposal points from Züri Fäscht 2019 (left) and 2023 (right).

In addition, specific guidelines described waste disposal for the feast and market stalls. In 2019 containers were available at each stand (emptied during the early cleaning). In 2023 the stands were assigned to a logistic point (LP) nearby, where they could bring their waste instead of having their own containers. 20 LPs were the bigger collection points, where waste was brought in from the EPs. They were distributed throughout the festival area and supervised by the ERZ employees. LPs contained a squeezing machine for mixed waste and containers for cardboard, mixed glass, PET and aluminium in bags. During operation, the stands separated their waste and collected it in their own containers or bags. Afterwards, the staff or the volunteering recycling heroes took these bags to the closest LP. The organizers planned to hire 1500 recycling heroes; however, only 400 registered. Their duties were to collect the trash and recyclables from the stands and the EPs and to bring them to the LPs. Before the recyclables were handed over to an LP, it was visually inspected to assess its level of purity. If the

recyclable content was above this 75%, it was categorized as pure and declared as recyclable. If it was below 75%, it was considered impure and disposed of for incineration.

In addition to the preliminary and subsequent planning, 350 ERZ employees and city cleaners were on duty during the three days and nights of the festival. The city cleaners were responsible for collecting litter from the ground and emptying the silver public trash receptacles, known as *litter sharks*, in use all year-round.

The disposal logistic departemnt of ERZ took the collected waste from the LPs to the respective recipients with waste collection vehicles on the road and boats on the river. As shown in Figure 5, the festival area was divided into sections numbered 1 to 4, each divided into subsections “a” to “f”. A detailed view on the right-hand side of Figure 5 also marks the locations of EPs, LPs, market, and feast stands.

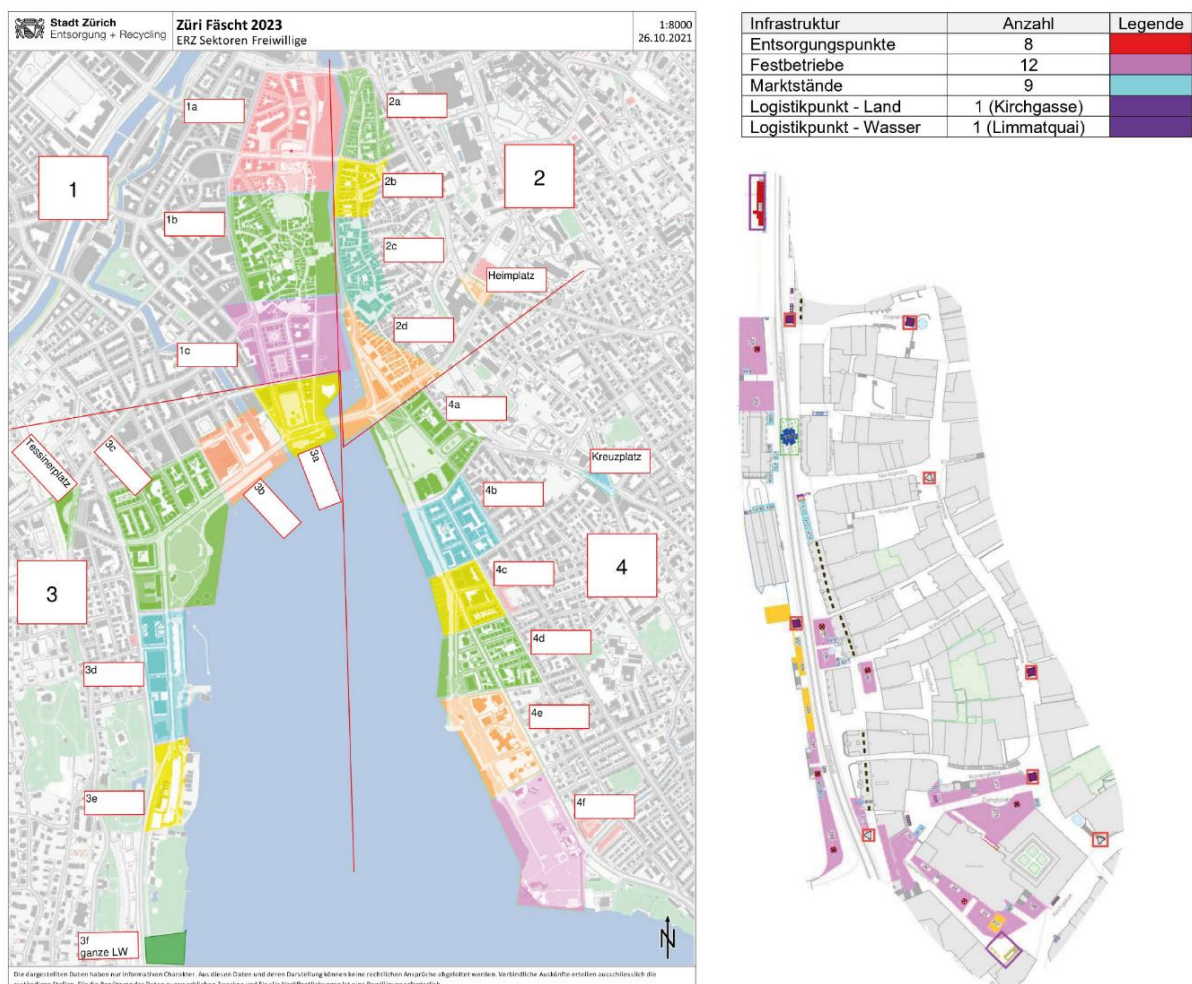


Figure 5. Layout of the sectors in the city (left) and a detailed view of area 2B (right).

3.1.2 After the festival

The waste collected from the ground was sent to the Kibag facility; a certain amount of the waste from the litter sharks - to Schnider AG; the material recovery facilities (MRF). Their manual and automated recovery methods sifts through the waste to recover additional recyclable materials. In Kibag, aluminum and PET are detected, whereas in Schnider AG only the manual sorting takes place for aluminum, PET, and glass. After, the sorted materials are sent to other facilities.

The materials, that got sorted at the festival, go different paths. PET and aluminum are collected by the recipient Trans Cycle itself. After separation, PET is sorted according to color and quantity, and pressed into bales. It is then sent to the processing center in Frauenfeld, where it is washed and shredded into flakes. Later, it is shipped to different companies, melted, and remanufactured to new bottles. Aluminum is pressed and sent for further recycling to an unknown facility. Cardboard and glass first goes back to Hagenholz and are later sent to the depot of Spross. After, the contractual partner, SBB, transports the goods further on their Cargo Railway system. Cardboard is recycled at Modul and Glass goes to Vetropack. The waste from the press-troughs goes to the incineration facility at the KVA in Hagenholz.

3.2 Qualitative Methods

This part of the thesis focuses on understanding the meaning and extent of the waste management system for Züri Fäscht. Participants' perspectives and opinion are captured in interviews and in the hands-on experience. Different festivals are compared to induce bigger picture about the global situation.

3.2.1 Interviews

This section, the rationale behind conducting various interviews is elucidated and the methodologies employed for these interviews are outlined. The interviews did not follow a question pattern and were not recorded. The notes were taken during the conversation or written down afterwards. The conversations were held in person and over the telephone. The collected opinions are incorporated in the evaluation.

Oli Beck was the main responsible for the waste management system at Züri Fäscht. He was a part of the planning and implementation group, challenged with realization of the festival's requirements with the resources available at ERZ. Thanks to his extensive experience in this area, he was a good contact person for specialist knowledge. With experiences from other Züri Fäscht events, he also served as a valuable reference. Other ERZ employees, such

as the waste collectors and the logistics managers, were also interviewed. Some conversations are not further mentioned in the manuscripts, but helped to acquire knowledge and complemented the picture of the WMS business.

One important conversation took place with Marius Inauen, the head of waste sorting from the sharks. His company had been involved in similar waste management initiatives for other festivals. Therefore, he possessed firsthand experience with the collected inventory and offered insights into the procedural aspects. In addition, telephone calls were made with Swiss Recycling industries. They offered insights into the national sustainability objectives and strategies for their integration within the industry, as well as ways to incentivize companies to adopt them.

3.2.2 Individual experience

The hands-on experience gave me a more holistic understanding of the variables and complexities that arose during implementation of the waste management concept. Though participation, I became a part of the waste management team at ERZ for a moment.

During the three-day festival, I was able to take two roles. I spent my first shift in the waste disposal logistics and the second shift as a recycling hero. Specific work clothing was required for both, i.e., pants, top, gloves, steel toe shoes, and a sun hat. During the work, I took photos and notes. I wrote down my experience in detail after the festival, as there was no time during the event.

First shift

The shift took place during the first night of Züri Fäscht and started at 1:20 a.m. The meeting point was at the ERZ headquarters in Hagenholz. The waste disposal logistics team consisted of almost 40 people. First, all employees gathered and were instructed by the two operations managers, Beck and Kölliker. Beck, who I accompanied, was main responsible for the left water basin, sectors 2 and 4. Kölliker supervised the right water basin. Every team member was assigned to a smaller team and their respective sector. Every team received a walkie talkie to communicate during operation.

For each sector, there was assigned at least one vehicle for PET and aluminium, one for glass, and one for mixed waste. Each of them needed one driver and two loaders. At 2 a.m. everyone drove from the headquarters to their respective sectors to assess the situation. Meanwhile, the operation managers surveyed the extent of the aftermath and helped solving unpredictable problems, such as: unemptied container from the shift before, wrong collected

PET/aluminium and non-functional press-troughs. Around 4:20 a.m., the festival area became increasingly empty. At 5 a.m., the police gave permission to begin the early cleaning. Meanwhile, the operation managers, Beck and Kölliker, met for an interim meeting and later continued to various LPs, back and forth. They drove along the streets to monitor their cleanness and the state of waste containers. All happened with time pressure to ensure that every street is clean and free for the main traffic to come.

At 8 a.m., the whole team met in the city for a final briefing. Afterwards, most of them drove back to Hagenholz. A few drove around again and emptied the forgotten dumps. Afterwards, I went home for lunch.

Second shift

During this shift I helped looking after the recycling heroes. On Sunday, we met in uniforms at 9 a.m. in Hagenholz and drove into the city. I was assigned to subsector 2b and gave the volunteers the instructions of their tasks. While they were collecting the trash, I was responsible for one LP by the river. When the containers were full of trash and recyclables, I sent a signal with my walkie talkie. Shortly afterwards, boats arrived and emptied the containers. The walkie talkie also helped in communication with other subsectors. Finally, in hot weather, the shift ended at 5 p.m. after a final debriefing.

3.2.3 Festivals in comparison

Züri Fäscht is obviously not the only festival dealing with environmental issues, as this problem occurs during every big gathering. The event's impact on the environment, including the use of energy, transportation, and resources, as well as large amounts of waste produced, is immense. For that reason, the sustainability concepts are important considerations for mass events.

Several events were contacted for interviews and researched online. Table 1 contains the results of the comparison. The festivals differ in many ways, e.g., size, location, and program. Most of the compared parameters are self-explanatory, other are briefly explained below. First, a closed area means that the event takes place in a gated zone with controlled doorway. An open area not fenced and not regulated nor controlled. Second, the online sustainability concept attempts to evaluate whether an organizer consciously deals with the topic of sustainability and communicates it online. Third, a depot system signifies a strategy for charging the consumers for every cup or cutlery with extra amount they get back after bringing the used items back

Fourth, the data availability for festival-generated waste assesses whether data is recorded, evaluated, and accessible online.

Table 1. Characterization of different festivals.

Festival	#visitors	Open/closed area	Location	Sustainability concept online	Depot system	Data availability	Duration [days]	Visitors/day
● good	● mentioned but not extensive	○ non-existing/small section						

3.3 Quantitative Methods

The quantitative methodology used in this thesis includes an analysis of key parameters of the waste management system. The following sections describe the approach used to assess the material flow of waste, the transport, the costs, the energy consumption, and the emissions. The assessment produces results for comparison and as a basis for decision making.

3.3.1 Material flow analysis

The Material Flow Analysis (MFA) is used to track the flow of materials through a system, in this case, the WMS of Züri Fäscht. This tool shows the movements, transformation, and the final destination of materials within a defined boundary. The inflows are the recyclables collected at the EPs, the waste collected from the ground and in the Eps, and the waste from the shark bins. The outflows are the quantities that get incinerated or sent for recycling. The analysis involves creating a mass balance equation for the system and states that the total input of material equals to the total output. The balance equation helps identify discrepancies, losses, and accumulation of materials within the system.

The Sankey diagram is a visual representation of the flow of materials. It illustrates the distribution and transformation of materials from one state to another. The states are characterized by the distinctive arrow-like, interconnected lines varying in width to represent the magnitude of the flow. The left side represents the inputs and the right side the outputs.

The Material recovery rate (MRR) is a metric used to measure the proportion of materials, successfully recovered, and recycled from the waste stream. It is defined as

$$MRR = \frac{\text{total amount of recyclables}}{\text{total amount of waste collected}} \times 100, \quad (1)$$

where the *total amount of recyclables* represents the quantity of materials that are collected and recycled; the *total amount of waste collected* is the overall amount of waste that is produced within a specific area and encompasses the recycled waste and the incinerated waste. Eq. (1) indicates the efficiency of a recycling program. Higher material recovery rate indicates that a

larger percentage of the waste stream is being diverted from disposal, contributing to reduced environmental impact.

Excel was used to build the Sankey diagrams and the tables for comparison between the festivals from 2023 and 2019. The values used in the diagrams and tables are sourced from ERZ and correspond to the officially communicated figures that have been disclosed to the press (Stadt Zürich, 2023a). It is important to mention that the amount declared as recycling does not account the loss factor. Instead, the collection rate directed toward recycling is reported, not the actual recycling rate.

3.3.2 Transport

The introduction of new process steps, such as sorting, more extensive recycling, or additional waste disposal, lead to increased transportation. The following analysis assesses the routes and examines how they differed from those used in the previous festival.

First, a list was drawn up and included all the destinations that were driven to from the time the waste was collected until it was incinerated or recycled. Bürkliplatz in Zürich was chosen as the initial destination, as it is in the middle of the four festival's sectors. The following paths of waste transportation were estimated through telephone interviews. Some distances had to be simply assumed as no information was available. After analysis of potential facilities, where the material could be sent to, the average distance of 45 km was estimated as a material transportation distance. Other destinations were located with maps and information from the Swiss recycling associations. The frequency of travel along a route and the associated conditions, including average speed, vehicle type, and truck load, were ambiguous. There was no precise information on the pollutant emissions of the vehicles, but clearly, the greater the distance, the more gasoline was required and, therefore, the more exhaust gas was produced.

The distance of the routes was measured in kilometers using Google Maps and recorded in a table for the years 2019 and 2023. Ultimately, the comparison involved the average distance covered per kilogram of waste.

3.3.3 Costs

The expenses of a waste management system arise from the investment in the infrastructure and its operation, including collecting and transportation. The costs of the waste treatment service, and the price of a particular material may vary depending on the current market situation. Information about material and personnel costs were recorded by ERZ itself

and communicated in confidence on request. In addition, figures communicated to the public were used as an additional source.

All costs are calculated in Swiss francs, CHF and are normalized with waste mass. Therefore, the years 2023 and 2019 are compared in CHF per kilogram of material.

3.3.4 Energy and emissions

Primary production is the raw material extraction and the manufacturing of a good. Secondary production involves the processing and transformation of materials. This includes manufacturing processes, where used materials are transported, sorted, washed, and reproduced. It is used as a synonym for recycling in this thesis. The Cumulative Energy Demand (CED) is a measure that assesses the overall energy consumption of a product or process. It accounts for the energy inputs and outputs through various stages of a product's life. This indicator is often expressed in energy units, such as gigajoules [GJ]. The Intergovernmental Panel on Climate Change, IPCC 2021, introduced a measure called GTP, the Global Temperature Potential, to quantify the greenhouse gas emissions. GTP100 estimates global temperature change at the end of a 100-years-long period. The impact of different greenhouse gases on global warming is expressed in tons of CO₂-equivalent.

EcoQuery v3.9.1 database was used to record the emissions generated and the energy required for waste management. CED and GTP100 were looked up for the primary production, secondary production, and incineration of each material: aluminium, cardboard, glass, and PET. Biowaste is neglected because of its small contribution and the lack of data. Subsequently, the values are normalized by the weight of material. The total energy and emissions produced and avoided by the festival are then compared for years 2023 and 2019. The avoided impact is the difference of the total amount of recyclables and is calculated as follows:

$$\text{Avoided impact} = \text{recycled} - (\text{incinerated} + \text{manufactured}) \quad (2)$$

The exact values from ecoQuery are listed in Table 11 and in the Appendix. As there is not enough data on cardboard, its values were supplemented with the data for paper, both made from the same raw material and produced in a similar way. However, the paper is bleached, and the cardboard has a corrugated layer. It was assumed that these two differences cancel out in the emission calculations. The data for glass refers to green only, as this is the one that can be produced with colour-mixed cullet. Material-specific values were found online and in the

work of McDougall et al., (2008). The emissions and energy data of the incineration facility from ERZ are also supplemented by the ecoQuery data.

4 Results and Discussion

In this chapter, both the qualitative and quantitative findings are introduced and analysed. The environmental consequences of integrating sustainable practices into the waste management system of Züri Fäscht are evaluated and discussed. The material flow is represented to give an overview of different operational streams. The transport routes and the costs are evaluated to depict the environmental impact of energy use and emissions. At the end, the specific case of Züri Fäscht is analysed in the global context.

4.1 Material flow analysis

Two Sankey diagrams represent the flow of waste collected in the last two editions of Züri Fäscht. 2019 is presented in Figure 6. Sankey diagram for waste collected from Züri Fäscht 2019 in tonnes.. The fluxes are straightforward; most of collected waste undergoes incineration with a small portion being recycled. Over half of the recycled material is glass., all the recyclables come from at source separation.

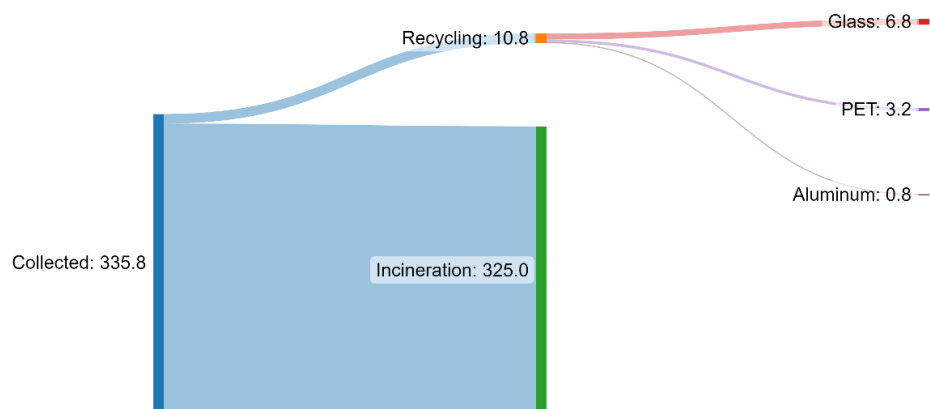


Figure 6. Sankey diagram for waste collected from Züri Fäscht 2019 in tonnes.

Figure 7 shows the material flow of the Züri Fäscht in the year 2023. The chart has a lot of dynamic and there are new stations in between the fluxes. Sorting 1 represents the Kibag facility, where all the waste from the ground gets sorted. Sorting 2 stands for the Schnider AG, who separates the waste from the shark bins. The total quantity of saved recyclables encompasses both the material collected at the EP/LP and the amount recovered from the cross-contamination of Sorting1 and Sorting 2. Overall, the recycling flow is smaller than the

incineration flux, but remains of representative size. Five distinct materials undergo recycling, with glass being the predominant contributor.

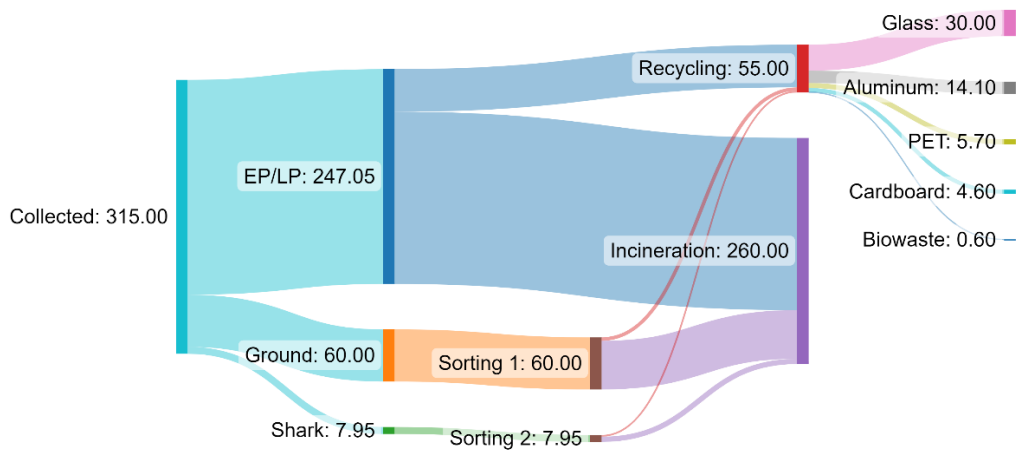


Figure 7. Sankey diagram for waste collected from Züri Fäscht 2023 in tonnes.

A closeup on materials saved at sorting facilities is presented in Figure 8. EP/LP could salvage the highest quantity of recyclables, amounting to 49 tons, constituting to nearly 90% of the entire recyclable material. Included in this number is the effect of the depot system and resulting correct separation at-stand. ERZ assumed that the waste directly sent from the EP/LP for incineration contains minimal to no recyclables, as both the stands and the visitors were able to dispose of it properly. The amount of material recycled from the sorting facilities was small. Initially, it was anticipated to send 120 tons of waste to Kibag (Sorting 1). Ultimately, only 60 tons of waste were found on the ground, a half of the anticipated amount. As mentioned, due to the technological limitations, Kibag could only sort two materials, aluminum and PET. The effort required for sorting using old machines is significantly higher, compared to the new machines, sorting more accurately and also detecting glass. Schnider AG (Sorting 2) was expected to receive up to 12 tons of waste, but only 7.95 tons were collected from the shark bins. The result was considered positive, three quarters was effective waste, such as the contaminated cardboard, wooden skewers, or leftover food. This again showed that the depot system and sorting at source is working efficiently. The pilot project with Sorting 2 was a first attempt, figuring out what the extent of lost recyclables is, which are still worth saving. Thus,

it was more for an inventory and less for the effect itself. In the future, the Kibag facility will be modified to detect a variety of materials and save possibly the most recyclables.

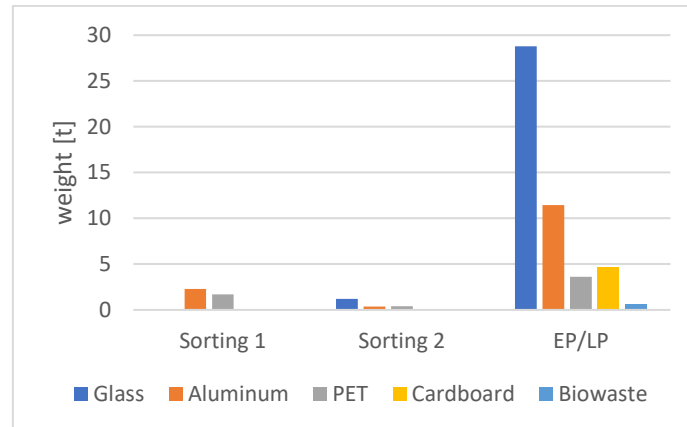


Figure 8. Materials saved from incineration at the different sorting facilities.

Five times more recyclables were recovered during the festival in 2023, compared to 2019. As mentioned earlier, glass was the most recycled material in both years per unit of weight. However, a 0.5 l glass bottle weights approximately 10 times more than a 0.5 l PET bottle. The number presented in this work are weight-based, not item-count-based. Therefore, the results do not provide exact information on the most frequently utilized packaging at the festival. Based on personal observations, cardboard appeared to be the most abundant, while glass was less visible. Plastic and aluminium cans were present in comparable quantities. Despite cardboard being frequently observed, it was also the most contaminated material, therefore, unusable. Aluminium has been resourced in a noteworthy quantity in 2023 compared to 2019, surpassing even the PET. Considering the material density, a 0.5 l PET bottle weights less than a 0.5 l aluminium can. However, during festivals, canned beer is consumed in large volume. PET bottles mostly contain sweet drinks for refreshment during the hot summer days and are used less at nights. The single-use cups that get handed out at the stands are not PET. Most of them are made from polypropylene (PP) and cannot be mixed. Overall, more plastic may have been disposed of, compared to aluminium, but ultimately only PET is recycled.

Controlling the input flow is challenging, especially considering it is a non-gated area. The reason why there was less waste collected in 2023, even with similar number of visitors, remains unclear. Since the number of visitors can only be estimated, there may well be deviations and the amount of waste per person would, therefore, be approximately the same. The waste flow going from collection straight to incineration is smaller in 2023, but still of a significant size. It could become smaller in two cases: if people would always dispose of the

recyclables correctly and if the technology at waste sorting facility becomes more efficient. The question on how to minimize the amount of incinerated waste still remains open. However, it is certain that at some point it is no longer sustainable due to financial and logistical costs. An illustration of this difficulty is determining the optimal quantity of EP and LP required, as opposed to merely serving as logistical overhead. What is also not known is how much and which valuable materials are still present in the flow between EP/LP and incineration.

The material recovery rates are listed in Table 2. The MRR for 2023 is five times bigger than for 2019, what is an overall success. Shark bins sorting has the largest MRR, compared to three different treatment facilities. This hand sorting is an effective strategy but needs a lot of time and effort. For big masses such as 315 tons of waste, it cannot be applied. Sorting at source has the second highest MRR. The earlier the material is isolated, the easier it is to restore it. Prevention is a crucial measure to avoid cross-contamination, ensuring that the material remains clean and in optimal condition for recycling. Sorting 1 performed the worst in terms of the MRR. The new process idea outlined above have the potential to change that.

Table 2. Material recovery rate.

	Recyclables [t]	Waste [t]	Waste collected [t]	MRR [%]
Total from 2019	10.8	325	335.8	3.23
Total from 2023	55	260	315	17.46
Sorting 1 from 2023	4	56	60	6.67
Sorting 2 from 2023	1.95	6	7.95	24.53
EP/LP from 2023	49.05	198	247.05	20.22

The benchmark set by the organizers, aiming for a 50% reduction in waste incineration, was not achieved. It is missed by 66 percentage points, i.e., 107.5 tons. However, this benchmark was rather unrealistic and can only be influenced by ERZ to a limited extent. 40% of total waste, equal to 126 tons, was produced just the last night. Logistically, this is a huge effort to clean in such a short time, the maximum utilization of logistical resources of ERZ, and the limit of its capacity. Change is needed in the overall festival operations. Learning from the waste hierarchy concept, prevention, and re-use have the biggest impact on waste generation but is not a part of ERZ's waste management responsibilities. It is a problem that need should be tackled at source. The feast and market stalls contribute largely to waste generation through their products and food packaging. This can be changed by implementing guidelines, such as the use of compostable dishes or the introduction of a depot system for dishes as well. Simultaneously, visitors also bring their own items obtained from supermarkets or their homes. The impact of this products can only be regulated by changing the overall rules,

such as the entrance control or drink limit per person. The question arises as to what society is willing to do or give up. Although, when examining the amount of waste generated per individual, the conclusions may be different. On average, each visitor produced 0.16 kg, as much as a medium sized apple. Therefore, another question emerges: how much more can this value decrease? In essence, waste is inevitably generated during every event. The key is transforming waste into a source of new input, fostering a life cycle.

4.2 Transport

The transportation routes from the Züri Fäscht 2019 and 2023 are represented in Table 3 and Table 4. Each route between the collection point and the recycling facility or incinerator is listed. In 2019, seven stations required driving, covering a total distance of 167 km. This year, the number of stations has increased to sixteen, totalling 487.6 km. Excluding Sorting 1 and 2, the remaining distance is 218.4 km, less than a half of the current total.

Determining the destinations of specific recyclables proved challenging. When inquiries were made, the information provided was often limited to generalities, such as glass going to Vetroswiss and aluminum to Igora. However, both entities, Vetroswiss and Igora, serve as intermediaries for advancing Swiss recycling rather than being the actual recycling plants. Unfortunately, these intermediaries could not offer details on the ultimate destinations of recyclables. To address it, attempts were made to identify nearby recycling centres online and calculate the distance. Estimations were required for the following destinations: Alu Menziken, Nussbaum Matzingen AG and Vetroswiss. Another impediment was tracking all the transits until the final station of the remanufactured good. This challenge is demonstrated on an example of PET. ERZ gave the information that PET will go to Trans Cycle for further treatment. The journey did not end there. At Trans Cycle, PET is simply compacted, after which it undergoes further processing at another recipient. The next stop is at Recypet, where it is washed and crushed into flakes. Subsequently, the majority travels to Germany for remanufacturing. Unfortunately, specific details about the location in Germany remain elusive and depend on the takers. This example underscores the extensive transportation required in the recycling process and the lack of transparency in the system. Another difficulty was to find the locations of various delivery stations, for instance Cargo SBB, were not straight forward. Additional details beyond the kilometers covered in the waste disposal process are omitted due to the lack of specific information about the vehicles used, the waste quantity per vehicle, or the operating hours. Furthermore, certain routes are travelled repeatedly and with multiple vehicles, a factor not considered in the total distance calculation.

Table 3. Routes of ZF 2019.

Routes	Distance [km]
KVA Hagenholz -> Zürich	7
Zürich -> KVA Hagenholz	7
KVA Hagenholz -> Trans Cycle	25.6
Trans Cycle -> RecyPET	63.6
Trans Cycle -> Igora (Alu Menziken)	34.8
KVA Hagenholz -> Spross/SBB	6.8
Spross/SBB -> Vetropack	22.2
TOTAL	167

Table 4. Routes of ZF 2023.

Routes	Distance [km]
KVA Hagenholz -> Zürich	7
Zürich -> KVA Hagenholz	7
Zürich -> Trans Cycle	32.3
Zürich -> Sorting 1	19
Zürich -> Sorting 2	87.1
Trans Cycle -> RecyPET	63.6
Trans Cycle -> Igora (Alu Menziken)	34.8
Zürich -> Spross/SBB	4.2
Spross/SBB -> Vetropack	22.2
Spross/SBB -> Model	47.3
Sorting 1 -> Trans Cycle	14.7
Sorting 1 -> KVA Hagenholz	9.4
Sorting 2 -> PRS Frauenfeld	46.3
Sorting 2 -> Igora (Nussbaum Matzingen AG)	41.7
Sorting 2 -> Vetroswiss	45
Sorting 2 -> KVA St. Gallen	6
TOTAL	487.6
TOTAL without Sorting 1 + 2	218.4

This Table 5 shows the distance traveled per kilogram of waste or recyclable. The total distance per kilogram of waste is three times greater this year than in 2019. Without the Sorting 1 and 2 facilities it would be twice as high. This increase is not surprising, as expanded recycling ideas such as additional saving of cardboard and organic waste also require more trips. The outlook for 2023 appears promising when considering the kilometer driven per kilogram of recyclables. Even if the Sorting 1 and 2 are included, the distance per recyclable is almost half as big as in the previous year. Without Sorting 1 and Sorting 2 it would be three times smaller than in 2019. Quantity does not always equate to improvement. Increased

logistical efforts should be approached with caution, especially as it often involves additional travel. For example, EP and LP contribute significantly to rescuing recyclable materials, but the associated logistics entail considerable effort. Transporting heavy metal grids for scaffolding requires time and resources. It would be valuable to explore the optimal number of EP and LP per area to minimize transportation needs. Other newly implemented ideas, like more frequent collections, have the potential to prevent material contamination but should be implemented in a judicious manner.

In the overall assessment, the Züri Fäscht 2023 result, excluding Sorting 1 + 2, demonstrates the best performance. This approach allows for the recovery of a substantial amount of recyclables while minimizing the traveled route. Ultimately, the sorting facilities did not save the materials, but the briefing with the festival stands, the presence during the festival and providing the sorting at source stations did. Transportation often gets underestimated even though it is one of the biggest sectors that cause emissions during the festival.

Table 5. Distance per kilogram waste collected.

	ZF 2019	ZF 2023	ZF 2023 without Sorting 1 + 2
Recyclables [t]	10.8	55	49.1
Total collected [t]	335.8	315	247.1
Distance [km]	167	487.6	218.4
[km/(kg recyclables)]	15.46	8.87	4.45
[km/(kg collected)]	0.50	1.55	0.88

4.3 Costs

Balancing the initial investment with the potential for reduced waste disposal costs requires careful planning. The budget for ERZ from 2019 was doubled for this year. The expenditures encompassed the following distribution: 43% external material expenses, 40% internal personnel expenses, 13% internal material expenses, 4% fee exemption. ERZ pays the sorting facilities a certain amount per tonne of waste. This can be expensive as there is no standardized process yet. The recyclables become their property and are resold at market price or for free. Some expenses are also unpredictable and arise afterwards, such as fees when there is too much waste or changes in the market price of goods. There are also certain costs that get covered. The PET and aluminum collected by Trans Cycle is paid by PRS, PET Recycling Swiss. Table 6 shows the costs per kilogram of material in 2019 and 2023. The material cost

per kilogram of waste collected was twice as high this year. At the same time, five times as many recyclables were saved, meaning that the price per recyclable material fell significantly.

Table 6. Total costs of the WMS during the festival.

	ZF 2019	ZF 2023
Budget [CHF]	470000	938097
[CHF/kg collected]	1.40	2.98
[CHF/kg recyclable]	43.52	17.06

Implementing innovative waste reduction and sustainability strategies involve upfront costs but can lead to long-term savings. The payoff in investing in Kibag's sorting system is contingent on the ability to save and resell enough recyclables in the long run. Some materials, like aluminum or green glass, can be more cost-effective to recycle because they retain their value in the market. Moreover, there is significant potential for high demand, given the strong promotion of the circular economy concept in Switzerland and its relevance to meeting climate goals. Incineration does also need several investments. Nowadays it is a common process and therefore cheaper. However, maintenance and regulatory requirements for emissions impact the costs strongly and can rise because of stricter regulations concerning the environment. Cost-effective waste management practices are essential for minimizing the environmental impact while ensuring the financial feasibility of the festival cleanup.

4.4 Energy and emissions

Figure 9 represents the overall energy impact of the collected waste for both years, including the contribution of each recyclable. Figure 10 does the same for the warming potential of CO₂. The resulting GJ and tons of CO₂-eq follows the same order as the weight of the collected material. Therefore, incineration accounts for the largest share of total energy impact and emissions. Both tables exhibit comparable alterations from 2019 to 2023, where this year's Züri Fäscht has a bigger CED but a smaller GTP100. This leads to the implication that recycling has a bigger energy impact but causes less emissions. Nevertheless, definite conclusions cannot be drawn, as there may be additional reasons and the accuracy of the data used is questionable. The amount of garbage from the Züri Fäscht cannot be recorded separately because it is burned with the remaining municipal waste and the information from ERZ are also annual average values and are not recorded specifically for one day. So, there may be a deviation because the composition of the waste at the festival differs compared to normal municipal waste. For example, there is certainly more cardboard in the trash because of the food stands. The data for the energy consumption and emissions is taken from a general

incineration plant with flue gas cleaning and not from the waste to energy plant of KV Hagenholz. The inclusion of transportation is also missing. The data for the recyclables does include the transportation between the different treatment facilities but it is just a rough global estimation. In conclusion, the conservative representation was because broad information of ecoQuery and non-covering or missing information from the actual actuators. Determining precise figures is challenging as each treatment method has its own characteristics and the system boundaries are defined in different ways.

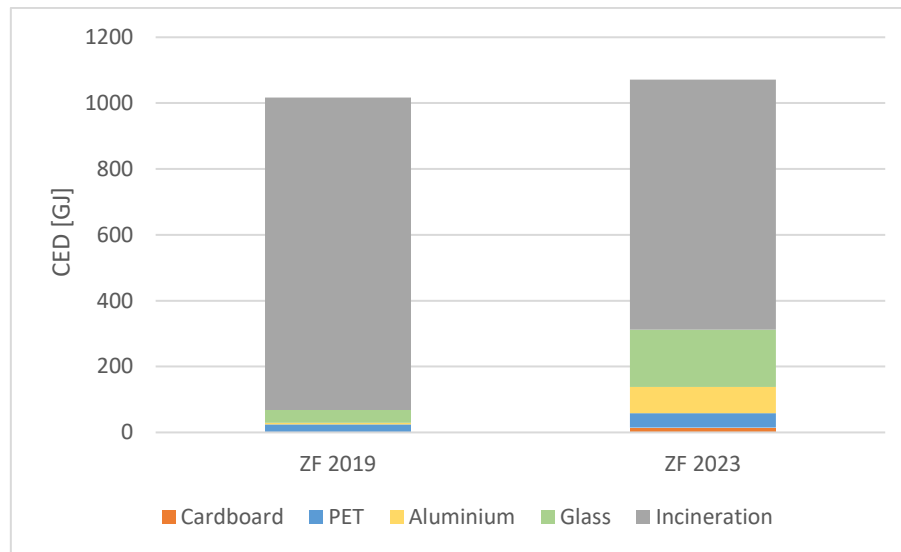


Figure 9. CED in comparison.

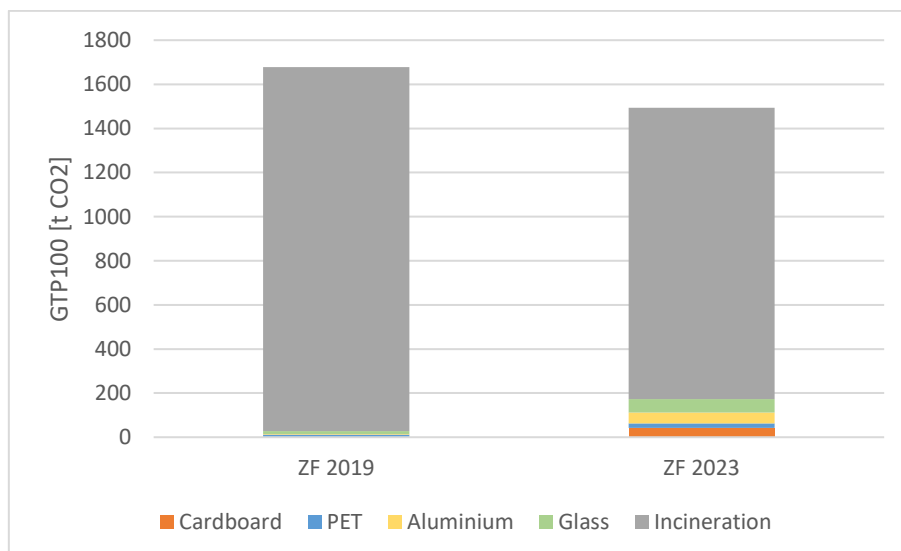


Figure 10. GTP100 in comparison.

The CED and GTP100 of the different recyclables are represented in Figure 11 and Figure 12. For each material the impact per ton of recycling, incineration and manufacturing is represented. According to that, aluminum, glass, and PET all have a positive impact when

being recycled rather than incinerated and newly manufactured. Recycled cardboard only has a positive effect on energy savings. Given the previously mentioned discrepancies, caution should also be exercised when interpreting the following diagrams. Therefore, less attention is paid to the exact numbers and more to the insights about the individual materials. The data for cardboard was hard to evaluate and needed to be substituted by paper. Reducing the extraction of virgin material can save biomass and prevent deforestation. But it also serves as an effective combustible, contributing to fossil fuel savings during the incineration process. So in certain instances, incineration may be environmentally advantageous compared to recycling, and caution should be exercised when assessing a particular case. Additionally, data can vary widely, depending on the type of paper and the consideration of virgin product extraction. The energy required to incinerate a PET bottle and produce a new one from crude oil exceeds the energy required to incinerate a PET bottle to various treatment facilities and remanufacture it. Therefore, recycling leads to a reduced environmental impact if compared to incineration and manufacturing virgin resources together. Aluminum is a non-ferrous metal and has a substantial reduction in environmental impact, when recycled. Both, aluminum and glass, have one of the highest recycling rates and can be reheated without compromising its properties. In the case of glass, proper sorting by type is also important.

The CED and the GTP100 are two measurements for evaluating the environmental impact but there are many more that can be taken into consideration, such as water emissions. Additionally, the evaluation is characterized by an overall uncertainty due to the difficulty in accurately defining the appropriate scope and the challenge in predicting long-term effects.

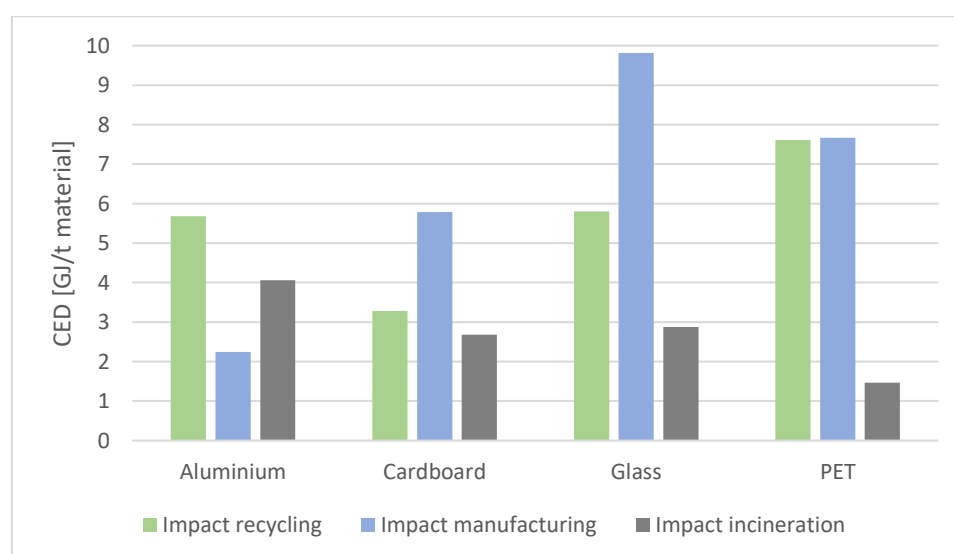


Figure 11. CED of different materials.

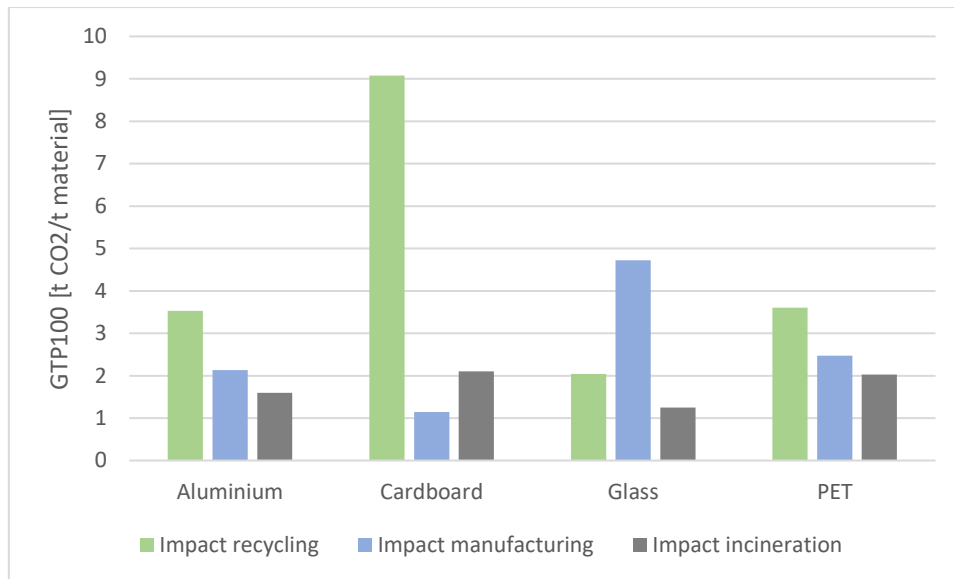


Figure 12. GTP100 of different materials.

On one hand, recycling prevents emissions and energy through material recovery. On the other hand, the waste to energy plant can also avoid emissions through heat and electricity recovery. How much the impact of both is, depends on the energy that was prevented. Table 7 is intended to give an idea of how much could be avoided through recycling at the Züri Fäscht 2019 and 2023.

Table 7. Impact avoided through recycling.

	CED avoided [GJ]	GTP100 avoided [t CO ₂ -eq]
ZF 2019	-52.24	-29.74
ZF 2023	-248.07	-98.98

On average, each attendee contributed to an emission of approximately 0.75 kg of CO₂-eq with just waste. This does not include the other sectors, such as transportation to the festival, food consumption or entertainment. In conclusion, Waste causes a lot of CED and GTP100, but overall it is still one of the smaller sectors that impact the environment during the festival.

4.5 Festivals in comparison

The major events under comparison are provided in the Appendix, and summarized in Table 8. There are fourteen festivals in total, and they all take place in Europe. Most of them take place in Switzerland, with notable international events including Tomorrowland in Belgium, Rock am Ring in Germany and Glastonbury in the UK. All the listed festivals and more were contacted but not everyone responded. The majority is situated within an enclosed area, with the location being either in the city itself or in a secluded rural setting. The benefit

of a closed area lines in the increased oversight of logistics and control over items entering and exiting the space. It facilitates a more comprehensive evaluation at the conclusion of the event, enabling a thorough assessment of the environmental impact. In contrast, an open area must adapt to the existing environment and has limited capacity to intervene in waste regulation. Being situated in an urban setting offers the advantage of utilizing pre-existing infrastructure and benefiting from enhances logistical facilities. However, obtaining permits in a city can be more intricate. In contrast, rural areas may lack existing infrastructure and can harm the natural habitat but provide an opportunity to define and control scalability. Hardly any online sustainability concepts were both comprehensive and highly informative. Many websites only feature brief articles on the topic, and locating them on the site can be challenging. There are often bullet point outlining their activities, yet detailed information about these activities is seldom provided. The majority has implemented a depot system. For those with a question mark, no information could be found online, and no reply was received when inquiries were made. This concept has only emerged on a large scale in the last few years and is now largely considered a standard practice. There are two concepts of depot system: one for reusable cups and one for single-use cups. Reusable cups can significantly reduce the volume of waste. As learned from the concept of the waste hierarchy, reuse is preferred over recycled single-use cups. But it is important to consider their lifespan and effort on how the reused cups get transported and treated afterwards. Implementing the depot of single-use cups can help reducing ground waste volume, thus positively impacting the logistical efficiency of the WMS. In general, a depot system conveys a positive environmental message, promoting awareness and contributing to a circular economy. There is rarely data available on how much waste everyone produces and what happens to it. Information on evaluating their sustainability concept is often not given. In response to a direct inquiry, they indicated that they do not do any review at all. Good examples for a transparent communication and work of reflection, about their caused emissions and the material flow, are the Glastonbury festival and the Winterthurer Musikfestwochen.

Table 8. General overview of different festivals.

Festival	#visitors	Open/closed area	Location	Sustainability concept online	Depot system	Data availability	Duration [days]	Visitors/day
ZüriFäscht	2'000'000	Open	City of Zurich	●	Yes	No	3	670'000
BigAir Chur	27'000	Closed	Coutryside, Chur	○	Yes	No	2	13'500
B-Sides	4000	Closed	Coutryside, suburb	◐	Yes	No	3	1'300
Churer Stadtfest	85'000	Open	City of Chur	○	Yes	No	3	28'300
Glastonbury	210'000	Closed	Coutryside, Farm	●	Yes	Yes	5	42'000
Luzerner Fasnacht	300'000	Open	City of Lucerne	○	No	No	4	75'000
Montreux Jazz Festival	250'000	Open & closed	City of Montreux	◐	Yes	Yes (on request)	16	15'600
OAFF	150'000	Closed	Nature reserve	◐	Yes	No	3	50'000
OpenAir SG	110'000	Closed	Nature reserve	●	Yes	No	4	27'500
Rock am Ring	90'000	Closed	Coutryside, suburb	○	?	No	3	30'000
Stolze OpenAir	8000	Closed (free entry)	Parc in Zurich	○	Yes	No	3	2'700
Streetparade	1'000'000	Open	City of Zurich	◐	Yes	No	1	1'000'000
Tomorrowland	600'000	Closed	Coutryside, suburb	○	?	No	6	70'000
Winterthurer Musikfestwochen	50'000	Open & closed	City of Winterthur	●	Yes	Yes	12	4'200
ZOA	50'000	Closed	Coutryside suburb	◐	Yes	No	5	10'000

● good ◐ mentioned but not extensive ○ non-existing/small section

The carnival in Lucerne has a long tradition and takes place in the whole city centre. The scope gets bigger every year, so does the waste problem. In 2019, a total of 120 tonnes of waste were collected and afterwards incinerated (Rüegger, 2019). Almost 10% of it lands in the river Reuss, which flows through the downtown area (Estermann, 2023). Sustainability measurements against this problem are small and an awareness concept does not exist. There is no information online about these topics. The city is in a state of exceptional circumstances, for which there is a certain tolerance. Many concepts are difficult to implement because of the unique conditions. For example, the depot system is hard to implement because there are a lot of diverse bars, wagons and unregistered stands that sell their own beverages. The waste on the ground is not resorted as urine and vomit heavily contaminate it. Important waste management measures are the distribution of the *Big Bags* in the whole area or the five-fold increase in the workforce of waste collectors (Regionaljournal Zentralschweiz, 2023). The aftermath is neither recorded, nor analysed through the administration of Lucerne. However, there is an exchange with the media, which reports during the big event and addresses the issue of waste, among other things. Benedikt Bucher, the responsible for the maintenance of the urban space, thinks it is difficult to reach the masses with a sustainability concept. Certainly, they are always open

for new ideas. For next year they plan new measurements concerning the waste avoidance on the river bank.

A good example is the music festival in Winterthur which plays downtown. On the website there is a comprehensive section about sustainability and their role in it. The main goal is a CO₂-neutral event. In order to get a better understanding of their emissions, they analysed and compiled their own eco-balance (Winterthurer Musikfestwochen, 2023). They worked with an experienced company and invested almost three hundred hours of work. This shows how complex and elaborate this analysis can be. According to their data, there has been 10.9 tonnes of waste produced in the year 2021 which is around 200 grams of waste per person. The emissions of the waste generation is small compared to other indicators. The transportation and gastronomy together account 80% of the total emissions (Winterthurer Musikfestwochen, 2021). Besides raising awareness and implementing different measures to dispose the waste eco-friendly, they also pay for the CO₂-emissions that occur. The organization team makes it clear that buying yourself into the climate neutral status is not the ideal solution. The minimization of emissions has first priority.

The Züri Fäscht ranks as the second largest in terms of daily attendance, surpassed only by the Streetparade. Both take place around the water basin and are open to the public. However, there are also some differences. The Streetparade, being a one-day event, is organized with a different logistical framework. While the Züri Fäscht necessitates continuous cleanup over a three-day duration, next to visitors and residents. The Streetparade's short timeframe presents challenges for implementing a depot system, particularly given the high percentage of attendees engaging in alcohol and substance consumption. These circumstances complicate the establishment of recycling stations and items wiped from the ground often lack recyclable materials, leading to sorting facilities no longer accepting such waste due to contamination and health concerns for the staff. The organizers acknowledge and promote the importance of waste disposal and saving recyclables on their website. They declare themselves as "the first major event in the world where 100% of all waste is separated and recycled" (Verein Street Parade Zürich, 2023). It shows that promoting sustainability at festivals has also become a trend. Supporting environmental concepts creates a positive brand image and can be financially beneficial by attracting sponsors. This economic incentive is not reprehensible, but the promises made are questionable and misleading.

Generally, there are a lot of ideas that some of the festivals do promote. The Zurich Openair stopped distributing free products in sampling size, OpenAir St. Gallen only sells Swiss meat, Glastonbury donates to environmental non-profit organizations and uses renewable energy, the Montreux Jazz Festival has organized a green team that sorts the trash and does a cleaning operation in the lake. It is difficult to perceive all the direct and indirect impact that a new idea can have. So, it is often based on a trial-and-error system. This emphasizes how helpful an exchange between the festivals can be. Conducting a self-assessment and making this information publicly available, can foster a greater understanding and learning experience.

Every festival listed in Table 8 has distinct baselines and constraints, varying in factors such as location, size, and entertainment. The great thing in common is the large accumulation of waste inner a concentrated area and within a short period of time. Addressing these environmental concerns and advocating for sustainable practices is crucial to mitigate their impact. There is no universal solution on how to implement this. Engaging with sustainability is both a privilege and a responsibility, which the festivals can afford.

5 Conclusion

This thesis revolves around the exploration of the viability of a sustainable waste management concept, considering the given logistical resources. The focus was on the material flow, transport, costs, and energy and emissions of the waste management system ERZ during the Züri Fäscht.

The results showed, the earlier the waste gets sorted the better the resources can be saved, contributing to the maintenance of a circular economy. Material that gets lost in waste is mostly not worth saving, due to cross-contamination and the extent of logistical effort. Transportation increased significantly as more recyclables were rescued and strategies, such as subsequent sorting were implemented. However, the potential of future sorting facilities is yet unknown and may be interesting to pursue. The recyclables themselves do not generate a large profit and sustainable waste-handling methods are often newer technologies that can cost more. Therefore, a certain amount of budget is required so that waste can be managed in an environmental-friendly manner while meeting the strict regulations. Statements about environmental impact must be considered with caution as there are often different technologies and scopes used for. An integrated waste management system includes both, recycling and incineration. Recycling primarily benefits by supporting the circular economy through material saving and averting CO₂ emissions that would occur during incineration and production. Simultaneously, incineration provides energy recovery. Finally, the comparison with other festivals highlights the varying levels of waste management systems and underscores the positive performance of ERZ.

Most Limitations arise from broad databases, rough estimations, and varying scopes. The quantities are rough projections and uncertainties is attributed to the diversity of information sources. Some used numbers are from newspapers, some are global averages, and some are specific from Switzerland but differ between facility treatments. Interesting information such as the composition of the waste was missing, which meant that the methods had to be adjusted several times. Additionally, the transparency of material flows and routes was missing, which made the assessment sometimes difficult.

Environmentally friendly waste disposal contributes to the sustainability of a festival, but it is only a small part of the big picture. There is no point in changing the output without first looking at the input. Waste mitigation lies also in the responsibility of the organizers and visitors. A few months after the festival in summer 2023, it was publicly announced that this

festival series would be discontinued. The end of Züri Fäscht shows that the actuators have reached their limits, and a rethinking of festival settings is necessary for the future.

6 Outlook

Based on limitations, there is often a lack of general data that are misleading and not useful to specific scenarios. Addressing this problem requires the establishment of a database featuring with well-defined framework conditions, including consistent measures of input and output streams. Knowing the composition of the waste would indicate which materials are most consumed and get lost. From this, new prevention strategies could be considered to embank waste generation. Another interesting task would be to rethink waste collection strategies in order to reduce logistical effort. Another approach could be the creation of a platform where festivals can access crucial information related to environmental awareness and engage in knowledge-sharing. It has the potential to facilitate the dissemination of expertise among festivals. Sustainability is also a marketing strategy and can be economic beneficial. Finding ideas on how to overcome the trade-off between environmental friendliness and cost-friendliness could help attract industries to invest in zero-waste management systems.

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Appendix

Table 9. Information source for festivals compared in this work.

Festival	Source
Züri Fäscht	(Züri Fäscht, 2023b)
BigAir Chur	(First Event AG, 2023a)
B-Sides	(B-Sides Festival 2023, 2023)
Churer Stadtfest	(OK Churer Fest, 2023)
Glastonbury	(Glastonbury Festival, 2023)
Luzerner Fasnacht	(Luzerner Tourismus, 2023)
Montreux Jazz Festival	(Fondation du Festival de Jazz, 2023)
OpenAir Frauenfeld	(First Event AG, 2023b)
OpenAir St.Gallen	(Openair St.Gallen, 2022)
Rock am Ring	(eventimpresents GmbH & Co. KG, 2023)
Stolze OpenAir	(Stolze Openair, 2023)
Streetparade	(Verein Street Parade Zürich, 2022)
Tomorrowland	(Tomorrowland, 2023)
Winterthurer Musikfestwochen	(Verein Winterthurer Musikfestwochen, 2023)
Zurich OpenAir	(Zürich Openair, 2023)

Table 10. Addresses used for transport calculations.

Destination	Adress
Zürich	Bürkliplatz, 8001 Zürich
KVA Hagenholz	KVA Hagenholz, Hagenholzstrasse 110, 8050 Zürich
KVA St. Gallen	KVA St.Gallen, Rechenwaldstrasse 30, 9014 St.Gallen
Sorting 1	Kibag, Moosackerstrasse 65, 8105 Regensdorf
Sorting 2	Schnider AG, Breitschachenstrasse 57, 9032 Engelburg
Trans Cycle	Trans Cycle, Ringstrasse 8, 5432 Neuenhof
RecyPET	RecyPET AG, Langfeldstrasse 80, 8500 Frauenfeld
Igora	unknown
vetroswiss	unknown
Spross/SBB	Spross Transport & Recycling AG, Hohlstrasse 330, 8004 Zürich
Vetropack	Vetropack AG, Schützenmattstrasse 48, 8180 Bülach
Model	Model Papierfabrik, Langackerstrasse 2, 5013 Niedergösgen
Alu Menziken AG	Alte Aarauerstrasse 11, 5734 Reinach (assumed)
Nussbaum Matzingen AG	Breitschachenstrasse 57, 9032 Engelburg (assumed)

Table 11. Information sources for estimating the energy and emissions.

	Aluminum	Cardboard	Glass (green)	PET	Incineration
Primary	China, (Wu Linlin, 2018)	Europe, (Hischier Roland, 2010b), Paper	Swiss, (Hischier Roland, 2010a)	Europe, (Fröhlich Thomas, 2018)	-
Secondary	Global, (Tur Sofia Parada, 2011)	Europe, (Hischier Roland, 2012), Paper	Swiss, (McDougall et al., 2008)	Swiss, (Kägi Thomas, 2017)	-
Incineration	Swiss, (Doka Gabor, 2010a)	Swiss, (Doka Gabor, 2010b)	Swiss, (Hischier Roland, 2010c)	Swiss, (Doka Gabor, 2007)	Swiss, (Stoikou Nikolia, 2023)



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